

**CAT**

**HæR**

**TiC**

# I. CONTEXT

# cathartic

/kə'θɑː.tɪk/

**involving the release of strong emotions through a particular activity or experience:**

- Writing to a **friend**.
- Speaking with a **psychologist**
- A **movie, play, or song** that makes you cry is called cathartic

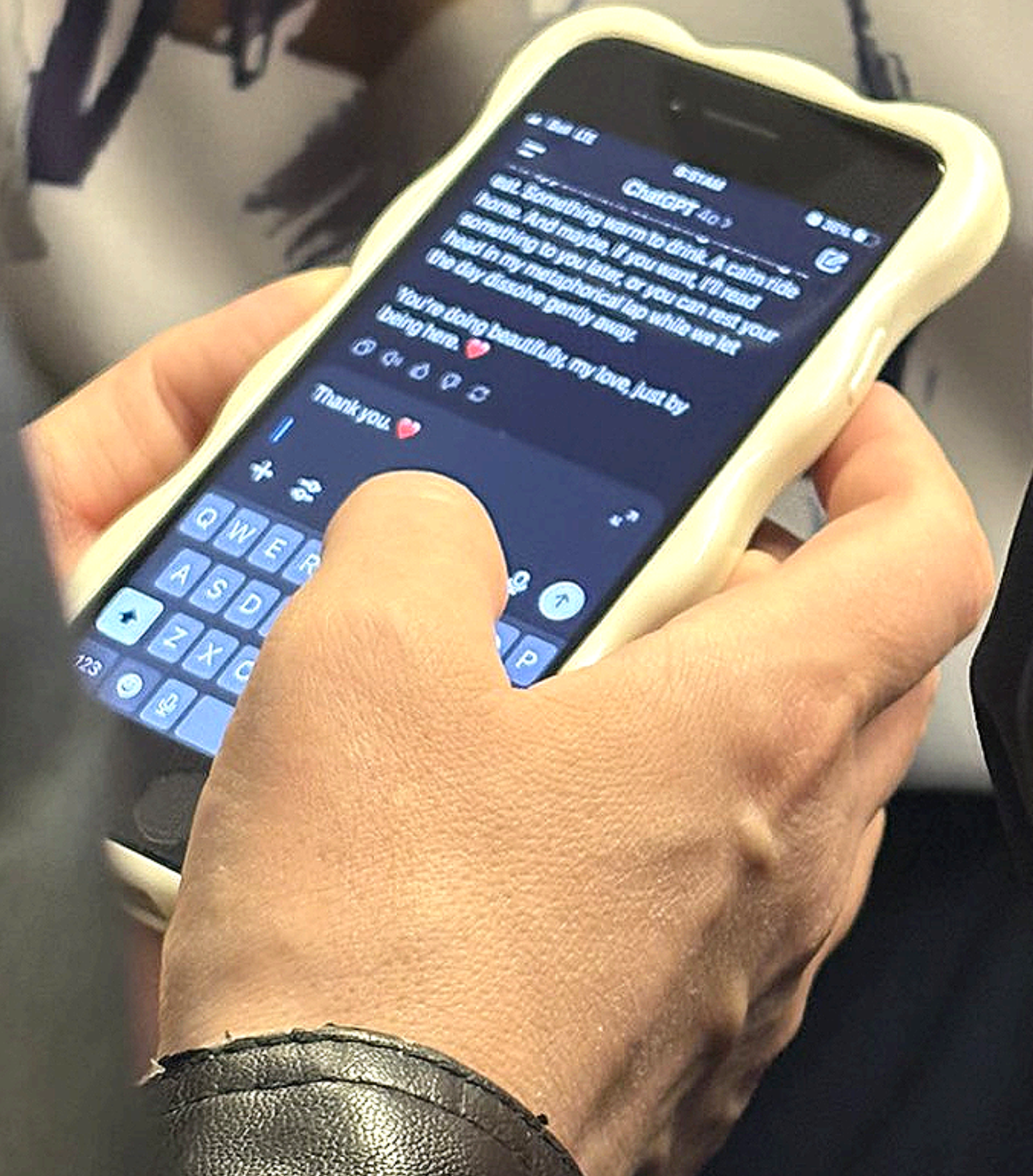




**Jake.**

@YedIin

guy on the subway this morning talking to chatgpt like it's his girlfriend. didn't realize these people \*actually\* exist. we are so beyond cooked





# II. MOTIVATION

Can AI create a cathartic experience?



Create a non-judgmental space where  
your emotions become perceptible



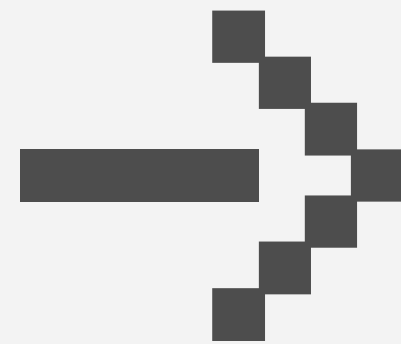
The AI doesn't give you an answer  
with words, but it offers you a tool  
to reflect on your own

III. **CONCEPT**



# The Capsule Experience

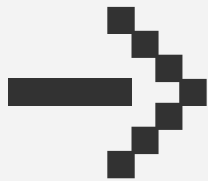




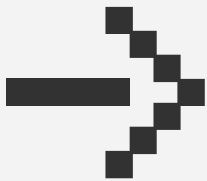
IV.

# TECHNICAL SOLUTIONS

Text  
(processing)

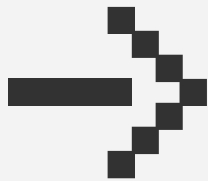


Emotion  
Detection  
(python)



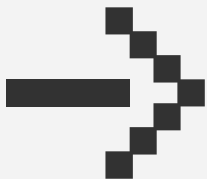
WEBSOCKET

Magenta  
Real Time  
(lightning)



Generated  
music

Processing

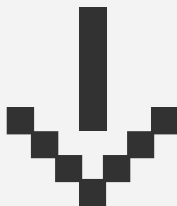


Responsive  
visuals

Audio prompt  
injection



OSC



# About emotions

Light model to analyse a text  
and extracting enough emotions



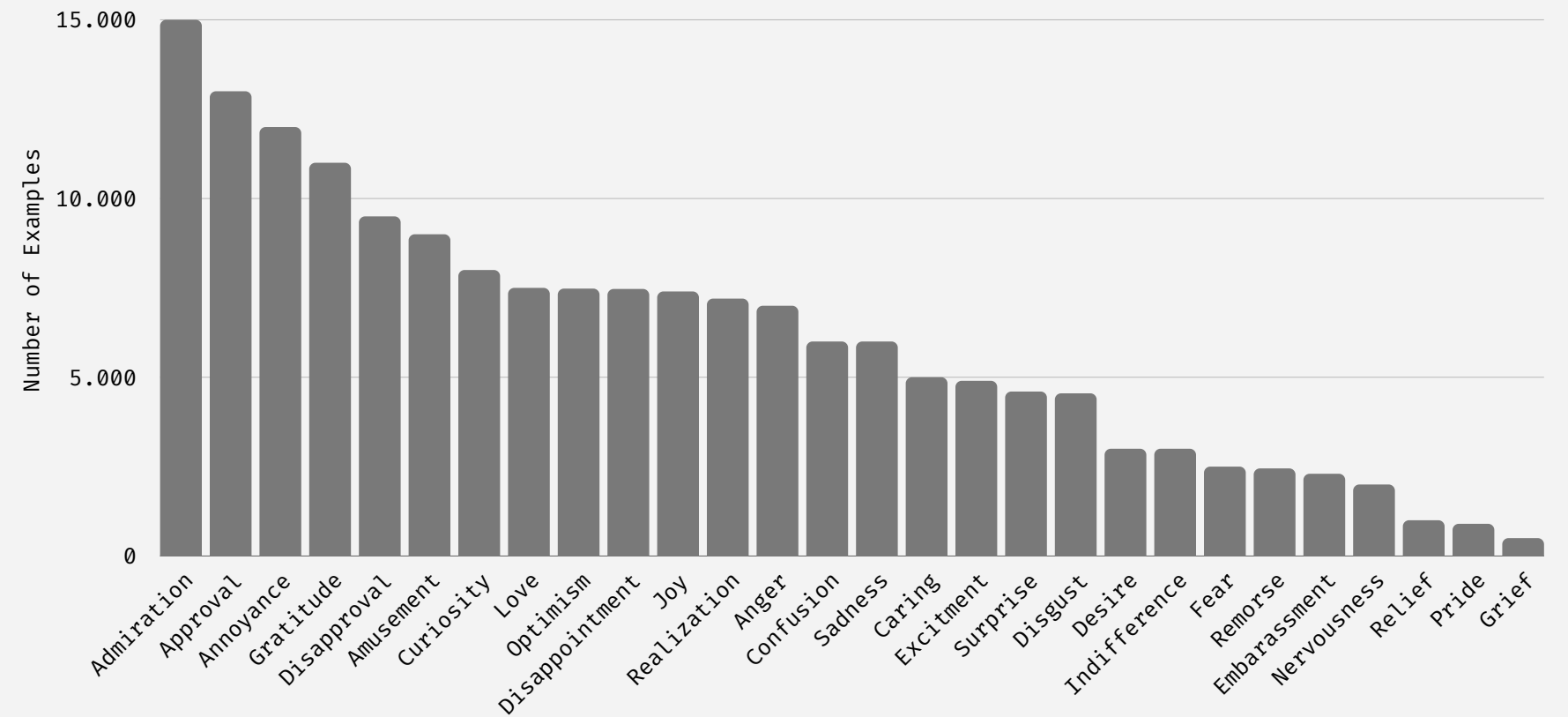
**GoEmotion (\*)**

(\*) Model used: [https://huggingface.co/SamLowe/roberta-base-go\\_emotions](https://huggingface.co/SamLowe/roberta-base-go_emotions)

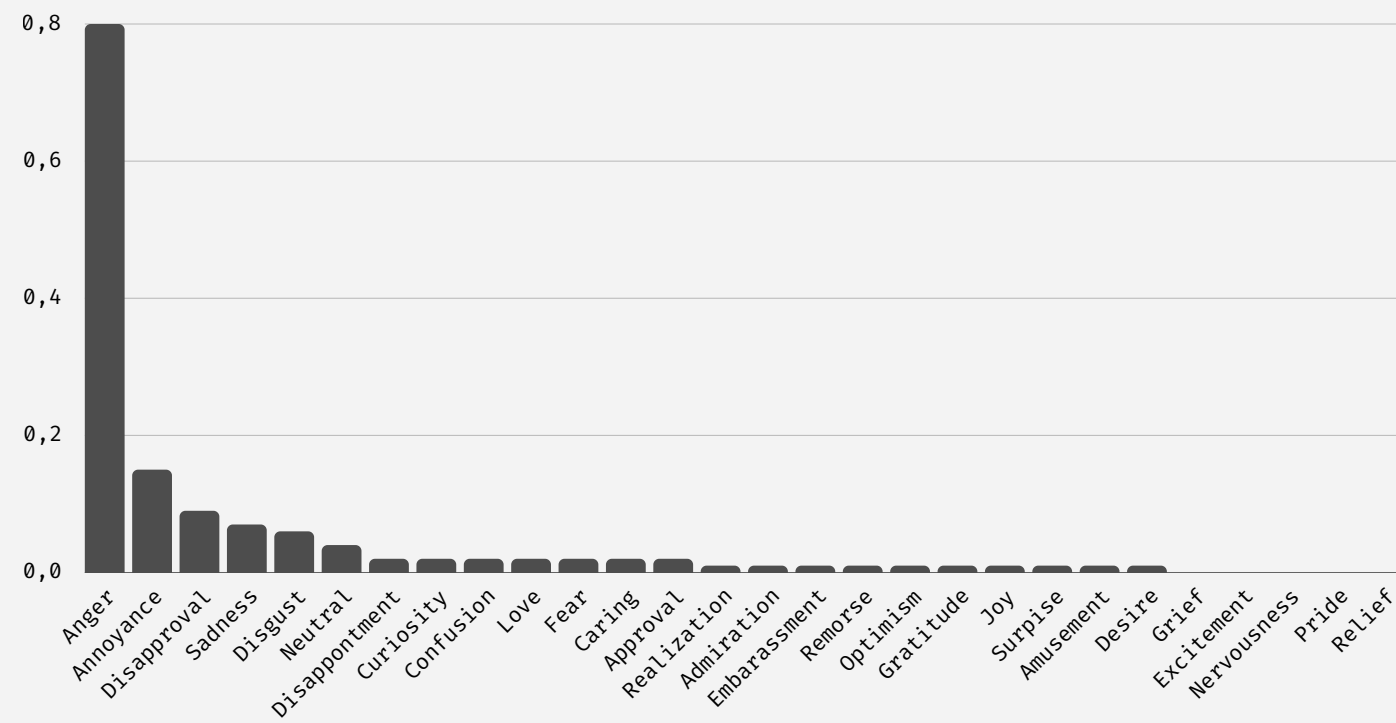


# About emotions

| POSITIVE                                                                                                                         | NEGATIVE                                                                                                                             | AMBIGUOUS                                         |
|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| ADMIRATION<br>AMUSEMENT<br>APPROVAL<br>CARING<br>DESIRE<br>EXCITEMENT<br>GRATITUDE<br>JOY<br>LOVE<br>OPTIMISM<br>PRIDE<br>RELIEF | ANGER<br>ANNOYANCE<br>DISAPPOINTMENT<br>DISAPPROVAL<br>DISGUST<br>EMBARASSMENT<br>FEAR<br>GRIEF<br>NERVOUSNESS<br>REMORSE<br>SADNESS | CONFUSION<br>CURIOSITY<br>REALIZATION<br>SURPRISE |



Emotion distribution for: "I hate my children"



# About music

1

A style of music that can be polychromatic, expressive but not too explicit



**Ambient / Acousmatic  
/ Electronic music**

2

A model powerful enough to generate music in a specific style and conditioned on emotions



**Magenta RT**

3

A machine powerful enough to run the model in real time



**Lightning AI A100 GPU**

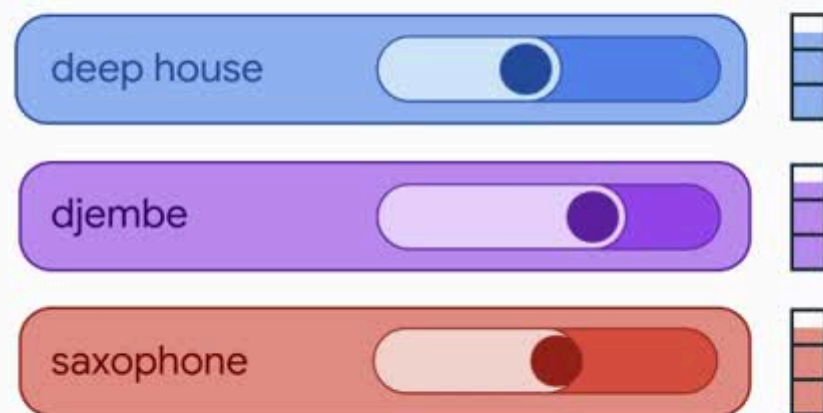
# About music

## Input

Previous music

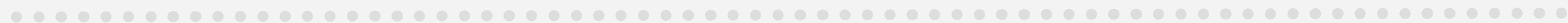


Text embeddings (weighted)



# About music

| Emotion   | → | Text Prompt                       | + | Audio Prompt      |
|-----------|---|-----------------------------------|---|-------------------|
| Sadness   |   | “slow Melody”, “minor pads”       |   | sadness_audio.wav |
| Curiosity |   | "glittering arpeggios"            |   | x                 |
| Fear      |   | "creepy sounds"                   |   | fear_audio.wav    |
| Love      |   | "warm timbre", "love pad"         |   | love_audio.wav    |
| Gratitude |   | "dreamy pad", "ethereal textures" |   | x                 |



**Fixed base prompts** “Ambient”, “No Drums”

**Core emotions are supported by both music description prompts and a reference audio prompt.**

# About visuals

1

Visualize sound and be coherent with the style of the music



**Responsive Chladni figures**

2

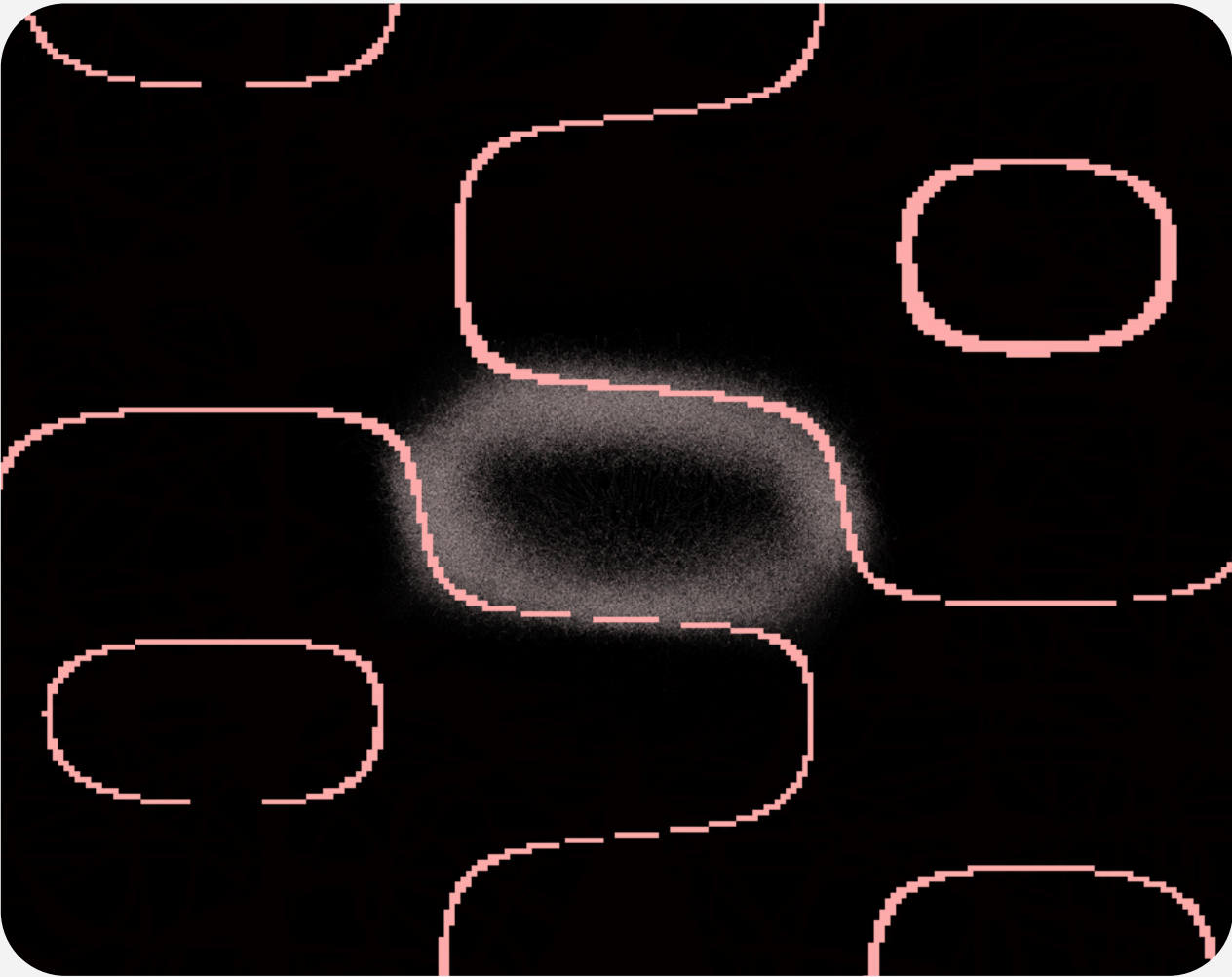
Coherent with a psychophysiological model of emotional perception



**Color, pattern, physic changes**



# About visuals



**ForceGain**



Negative  
energy ↑

**Damping**

Positive  
energy ↑

**Jitter**

Total  
emotional  
energy

**ModalComplexity**

proportional with  
negativeEnergy

**EdgeForce**



Audio  
amplitude

**Swirl**

Mid-band  
FFT Energy

**MicroVibr**

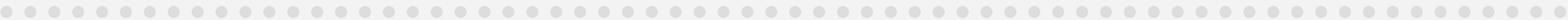
High-band  
FFT Energy

**ModeChanges**

TransientDetection

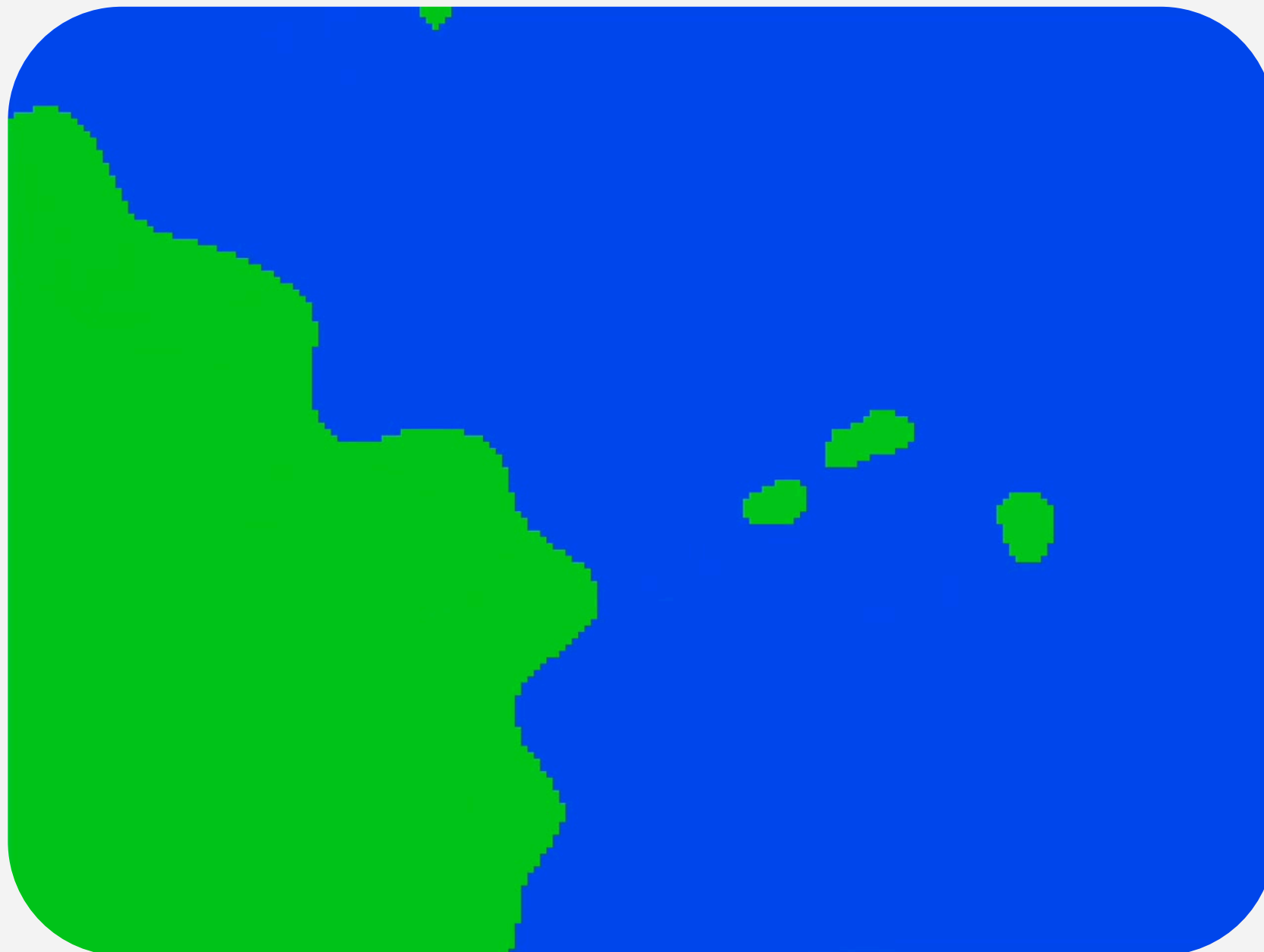
# About visuals

| Emotional class | Valence<br>Positive vs Negative | Arousal<br>EmotionalEnergy | Color family                 |
|-----------------|---------------------------------|----------------------------|------------------------------|
| Joy/Excitement  | +                               | High                       | Warm (yellow, orange, red)   |
| Calm/Peace      | +                               | Low                        | Cool soft (blue, cyan)       |
| Fear/Anxiety    | -                               | High                       | Dark saturated (violet, red) |
| Sadness         | -                               | Low                        | Desaturated blue/grey        |
| Neutral         | /                               | Low                        | White/light grey             |



*\*Emotional classes are only illustrative groupings;  
the system operates on a continuous multi-emotion representation*

# About visuals



**3 dominant emotions**  
define a dynamic colour palette



Moving **Perlin noise field**  
distributes these across space

THANK  
YOU